MIL-B-21741C(SHIPS)
14 June 1965
SUPERSEDING
MIL-B-21741B(SHIPS)
1 July 1963

MILITARY SPECIFICATION

BOOK, TECHNICAL, MAINTENANCE STANDARDS

1. SCOPE

1.1 Coverage. - This specification sets forth the requirements for preparing Maintenance Standards Books, and revisions thereto, for ship or shore-based electronic equipment.

1.2 POMSEE. - The term "POMSEE" is the abbreviation for Performance Operational and Maintenance Standards for Electronic Equipment. It is a program designed to assist Navy civilian and military personnel in establishing standards of performance and maintenance for, and the checking out of, electronic equipment. **It is a joint endeavor of the Bureau of Ships, Shore Facilities and Forces Afloat which are cognizant over the installation and maintenance of these equipments.**

1.3 Maintenance Standards Books. - Maintenance Standards Books are the fundamental elements of POMSEE. They provide the test procedures for determining standards of performance and maintenance of the equipments or systems for which they have been prepared. In addition, the books provide contractors' recommended periodic preventive maintenance schedules.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-M-15071 - Manuals, Equipment and Systems.
MIL-L-17192 - Lubrication Design, Lubricants, and Lubrication Information for Electronic Equipment; General Specification.
MIL-S-21740 - Sheet, Technical Data, Performance Standards

STANDARDS

MILITARY

MIL-STD-415 - Test Points and Test Facilities for Electronic System and Associated Equipment, Design Standard for.

PUBLICATIONS

DEPARTMENT OF DEFENSE

DD Form 441 (Attachment) - Industrial Security Manual for Safeguarding Classified Information.

FSC 7610

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T10:26Z
Check the source to verify that this is the current version before use.

BUREAU OF SHIPS

NAVSHIPS 91727 - Electronic Test Equipment Application Guide.

NAVSHIPS 94324 - Maintainability Design Criteria Handbook for Designers of Shipboard Electronic Equipment.

NAVSHIPS 250-660-42 - Electric Shock - Its Causes and Its Prevention.

(Copies of specifications, standards, drawings, and publications required for suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Definitions (See 6.2).

3.2 Level of writing. - The procedures shall be written to a level no higher than the comprehension level of a high school graduate having had experience in the operation and maintenance of the equipment or of similar equipment and in the general use of related test equipment.

3.3 Contents.

3.3.1 Material Content. - Maintenance Standards Books shall contain the following types of data:

(a) Front matter, consisting of such items as a Reference Standards Summary Sheet, title page, list of effective pages, equipment functional block diagram, list of test equipment requirements, list of estimated time to perform the tests, and a list of the special procedures required to assure accurate results.

(b) A series of detailed procedures for making the necessary tests to establish reference standard values by the installing activity for a specific equipment or system and to be followed by maintenance personel periodically to ascertain whether that equipment or system is performing satisfactorily. Included are also the limits of reference standard values, initially established by the contractor during production tests, within which measurements on a specific equipment or system must fall to be acceptable. These will provide the reference values indicative of normal operation and which are referred to during preventive or corrective maintenance to determine whether a circuit or equipment function is operating normally. Also included are electromechanical preventive maintenance procedures considered necessary by the contractor to ensure satisfactory operation of the equipment or system.

(c) A list of those reference standards tests and electromechanical procedures recommended by the contractor for periodic maintenance.

3.3.2 Style. - The style of contents shall be as illustrated in appendix 1.

3.3.2.1 Publication constants. - Running heads and feet for all pages shall be as illustrated in appendix 1.

3.3.2.2 Page numbering. - As illustrated in appendix 1, text pages shall be numbered with Arabic numbers, starting with "0". Front matter pages shall be numbered

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T10:26Z
Check the source to verify that this is the current version before use.

in lower case roman numerals. The title page (see 3.3.3.1.3) shall be assigned the numeral "i" but shall not be numbered.

3.3.2.3 Typography of reproducible copy. - Typography shall be of the cold-composition process. Except for the front cover and title page (see appendix 1), ten-point book-face type shall be used. Larger book-face type reduced to ten-point size will be acceptable if no increase in cost results. The right-hand margin shall not be justified (aligned).

3.3.2.3.1 Front cover and title page. - Roman type face shall be used for all data shown in appendix 1.

3.3.3 Form. - The form of the contents shall be equivalent to that illustrated in appendix 1.

3.3.3.1 Front matter. - Unless otherwise specified herein, the front matter shall conform to MIL-M-15071, and shall be included in the order indicated below.

3.3.3.1.1 Covers. - The covers shall contain the information shown in appendix 1, except as follows:

(a) If the book is unclassified, the word "UNCLASSIFIED" shall be printed at the top- and bottom-center of the covers in lieu of the security classification, and the security notice shall be omitted.
(b) If the book relates to only one model of equipment, "OF MODEL _____" shall be omitted.

3.3.3.1.2 Reference Standards Summary sheet. - One Reference Standards Summary sheet (see 6.2.7) shall be attached permanently in the front of each Maintenance Standards Book, immediately following the front cover. The Reference Standards Summary sheet shall provide spaces for recording the results of all reference standards tests. Each space shall be identified by the step number which provides the instructions for accomplishment. Each space shall contain the respective reference standard unit of measurement (e.g., amp, db, or v). When applicable, waveforms shall also be included to show the points on the pattern where the measurement is to be taken. In addition, critical installation data shall be included, such as "length of transmission line". As shown in appendix 1, the sheets shall be page numbered by consecutive Arabic numbers, starting with "1" and shall indicate the total number of pages (e.g., page 1 of 1, page 3 of 5, etc).

3.3.3.1.3 Title page. - The title page shall contain the information shown in appendix 1, except as follows:

(a) If the book is unclassified, the word "UNCLASSIFIED" shall be printed at the top- and bottom-center of the page in lieu of the security classification, and the automatic, time-phased downgrading and de-classification notice shall be omitted (see 3.3.4).
(b) If the book relates to only one model of equipment, "OF MODEL _____" shall be omitted.

3.3.3.1.4 List of effective pages. - The list of effective pages shall list all pages of the manual and shall indicate the issue information of each page. In addition, this page shall contain the name and address of the contractor and the contract number. As shown in appendix 1, a book error-correction statement shall also be included.

3.3.3.1.5 Field changes and revisions. - The field changes and revisions page shall contain the information shown in appendix 1. All field changes in effect shall be considered in developing the Maintenance Standards Book. As shown in appendix 1, the field

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T10:26Z
Check the source to verify that this is the current version before use.

changes considered in preparing the book shall be entered above the change record. If no field changes are in effect, this fact shall be so stated. In addition, tabular space shall be provided for recording the effects on the results of any step caused by field changes issued subsequent to publication.

3.3.3.1.6 Introduction. - The introduction shall contain only information necessary to understand the basic concept and use of the book, and shall be equivalent to that shown in appendix 1.

3.3.3.1.7 Test parameters, test equipment, and special tools. - Test equipment, test adapters, special tools, and materials which are required to accomplish the procedures contained in the Maintenance Standards Book shall be listed, as shown in appendix 1: reference shall be made to the recommended maintenance periods in which the equipment is used. In addition, in order that a suitable test equipment may be substituted by the technician or operator when the listed equipment is not available, the test parameters and the required accuracies for the measurements to be made shall be given in tabular form.

If it is impossible to accomplish a required procedure without assembly of a special jig or apparatus, sufficient construction details shall be given the Bureau or agency concerned prior to preparation of the manuscript, so that the Navy can determine whether the item shall be manufactured and distributed through supply channels or whether instructions shall be given for fabrication in the field.

When a test equipment, special jig, or apparatus is required only for accomplishment of the reference standards tests by the installing or overhaul activity, and is therefore not needed by the operating activity, this fact shall be so stated.

In general, the requirement for a special jig or apparatus to accomplish a required procedure will be avoided whenever possible.

3.3.3.1.7.1 Selection of test equipment. - General-purpose test equipment shall not be specified when suitable equipment test facilities (see 6.2.11) are provided. When general-purpose test equipment is required, the test equipment shall be selected from the latest edition of the Electronic Test Equipment Application Guide, NAVSHIPS 91727. (Copies of this publication may be obtained at the office of the Government inspector.) When a test equipment not listed in NAVSHIPS 91727 is required, prior approval for its use shall be requested from the Bureau or agency concerned. When approval is requested, the parameters of the required tests together with required accuracies shall be delineated, and the nomenclature, description, and characteristics of the proposed test equipment shall be given so that the test equipment may either be obtained by the Navy or an appropriate substitute specified.

3.3.3.1.8 Summary of estimated time to perform POMSEE. - As shown in appendix 1, the estimated time required to perform the scheduled maintenance shall be summarized to provide the following information:

(a) An estimate of the time required to accomplish each scheduled procedure. The estimate is to be based on the assumption that the test is to be carried out by an operator or technician familiar with the prime equipment under test. Where external test equipment is required to perform the check, the estimate of time shall be based on the availability of test equipment at the location of test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T10:26Z
Check the source to verify that this is the current version before use.

3.3.3.1.9 Special procedures. - Any special procedures or adjustments which must be followed in order to obtain accurate test results shall be included. These usually include detailed procedures which are prerequisite to more than one maintenance procedure. Corrective maintenance adjustments and procedures, such as i-f alignment and discriminator adjustment which are included in the equipment technical manual, shall not be included.

3.3.3.1.10 Block diagram. - As shown in appendix 1, a block diagram identifying the major functional sections of the equipment or system shall be included. Each major block shall be identified by its functional designation and a letter. This identification serves to establish the basic arrangement for the book (see 3.3.3.2).

3.3.3.2 Text pages. - The text pages shall contain all procedural instructions, illustrations, if needed, waveforms, and charts necessary for accomplishing the reference standards tests and for performing other preventive electromechanical maintenance on the equipment or system. The pages shall be arranged by functional sections which agree with the functional sections of the block diagram (see 3.3.3.1.10) and shall be followed by the contractors list of recommended periodic maintenance checks.

3.3.3.2.1 Reference standard tests (see 6.2.5). - These procedures, which are utilized in initially establishing reference standard values for a given equipment or system, are also used in subsequently carrying out preventive maintenance and performance monitoring routines on a periodic basic. Only such tests which provide quantitative measurements and which result in significant standards that have a direct bearing on operational performance and preventive maintenance shall be selected. Limits of the required values shall be provided by the Contractor for each reference standard test in order to ascertain acceptability or non-acceptability of initial values obtained on a specific equipment or system by the installing activity. These limits also serve in assessing performance during periodic maintenance of the equipment or system. These limits shall be established initially by the Contractor during production tests and indicated in the "Reference Standard" column of each test.

3.3.3.2.2 Preventive electromechanical procedures. - These procedures shall include all electromechanical checks considered by the Contractor to be essential in carrying out effective periodic maintenance on the equipment or system. They shall include specific mechanical or electrical maintenance and inspection instructions (e.g., lubricate antenna drive gear, clean modulator air filter, check cooling water levels, clean pedestal oil filter, inspect for proper oil level in antenna gear box, and burnish commutator and slip-ring brushes).

Note. - Do not include such preventive-maintenance procedures as tube testing, general cleaning practices, and common-sense inspections.

3.3.3.2.3 Procedural instructions. - As shown in appendix 1, the instructions for accomplishing the reference standards tests and preventive-maintenance procedures shall be clearly stated and shall conform to the following requirements:

(a) Instructional pages shall consist of general test requirements and one or more procedural tables (see 3.3.3.2.3.2) containing narrative instructions, waveforms, and illustrations, when necessary, arranged in such a manner that all controls, adjustments, test equipment, test points, and connections

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T10:26Z
Check the source to verify that this is the current version before use.

which require use in performing the steps may be associated readily.

Note: - When the narrative instructions for a given step are clear and simple, and where an illustration will serve no useful purpose, an illustration is not to be included.

(b) The instructional pages shall be divided into functional sections with all tests in each functional section arranged in such a way that they may be accomplished in logical sequence.

(c) Insofar as possible, the steps within each functional section shall be arranged by major units and grouped to facilitate the checking of all units in a given area.

(d) When space permits, procedural tables and illustrations, if included, shall be arranged on a single page. When the test information cannot be confined to a single page, all related test instructions and illustrations shall be restricted to two facing pages.

(e) When a single illustration is used to illustrate particulars for more than one step, the portion of the illustration pertaining to each separate step shall be identified by the pertinent "step number", which shall be enclosed in a circle or star, as applicable (see 3.3.3.2.3.2(a)).

(f) The preventive-maintenance aspects of mechanical and electromechanical items, such as motors, generators, dehydrators, pumps, cooling systems, antennas, amplifiers, and coupling units, shall be clearly detailed.

(g) The names of all parts which are titled or labeled on the equipment shall be capitalized and spelled exactly as they appear thereon.

(h) Except when referring to a front panel or other easily identified control, the reference designation (enclosed in parentheses) shall follow the part name; for example, BIAS ADJ (1R201). If the control or part is identified only by a reference designation, then the functional name shall precede the designation; for example, Panel Meter (3A2M2).

3.3.3.2.3.1 General requirements. - As shown in appendix 1, general requirements that apply to each step in the procedure tables shall be given under the heading "Operating Conditions and Control Settings". The general requirements shall conform to the following:

(a) If a switch, dial, or other control is to be specified in a certain position, its name and position shall be in all capitals (except numerals) and separated by a colon; for example, RANGE MARKER switch: 5.

(b) If an equipment or unit condition is to be specified, a direct statement defining the condition shall be made; for example, Radar Set AN/SPQ-5 in Acquisition Mode.

(c) If it is necessary to change the operating conditions and control setting for a given step, this shall be indicated within the step and shall pertain only to that step.

(d) New general control settings or operating conditions may be included between steps. Unless otherwise noted, these new conditions shall apply to all succeeding steps on the page.

3.3.3.2.3.2 Procedural tables. - Procedural tables shall conform to the most suit-

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T10:26Z
Check the source to verify that this is the current version before use.

able treatment illustrated in appendix 1; the specific treatment selected shall provide a clear presentation and economical utilization of space. The procedural tables shall be divided into columns, as follows:

(a) "Step No." - This column shall include a step number for each given test or procedure. Except for tests that are included in the Performance Standards Sheet for the equipment (see 6.2.9), each number shall be enclosed by a circle or ellipse; each step number for Performance Standards Sheet tests shall be enclosed by a star. The first procedure in each functional section shall start with step number "1", and the number shall carry a prefix letter denoting the functional section (e.g., A1 , A2 , and A3 for section A, steps 1, 2, and 3, respectively). All following procedures shall progress sequentially throughout the section. Decimal increments (e.g., 1.1, 1.2, and 1.3) may be used for substeps. When the book covers more than one model of equipment, letter suffixes (e.g., A1A , A1B , and A1C) may be used to identify those steps that are peculiar to particular models of the equipment or system.

(b) "Action Required." - This column shall include a statement which describes briefly the nature of the procedure to be accomplished.

(c) "Procedure." - This column (or space, depending upon the format of the table; see appendix 1) shall include complete instructions and data regarding any action other than that called for in the general requirements. If a step is affected by a production change or field change that is not to be incorporated in the entire equipment population, give separate instructions concerning what action should be taken if the equipment change has not been accomplished. Indicate the exact hookup of test equipment point to point, thus eliminating the the requirement for illustrations. Also indicate that all controls are to be returned to their normal positions upon completion of test.

(d) "Read Indication On." - This column shall be included only when a measurement (reading) is to be obtained. The column shall indicate the test equipment or functional designation and reference designation of the test set, indicator, dial, or meter, or any other information necessary to identify the item or part from which the measurement is obtained.

(e) "Reference Standard." - This column shall be included only with steps for which a reference standard is to be entered. The spaces for recording the standards shall indicate the unit of measurement and the maximum and minimum limits that can be expected from a normal measurement.

3.3.3.2.3.3 Illustrations (MAY BE OMITTED EXCEPT FOR COMPLICATED TESTS AND LUBRICATION PROCEDURES). - Illustrations, when required, shall be line artwork prepared with line weights of sufficient strength to reproduce sharply and clearly. In addition, illustrations shall clearly locate the applicable test points, test equipment connections, control settings, adjustments, lubrication points, and equipment parts necessary to aid in the performance of the steps listed in the associated procedural tables. Illustrations shall incorporate the details listed below.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T10:26Z
Check the source to verify that this is the current version before use.

(a) Controls and test points. - Titles shall be provided to identify all pertinent controls and test points. When necessary for clarity, the applicable test point, control, or part shall be outlined (in a rectangle or circle) on the view of the major unit or chassis, with a leader or guideline leading to an enlarged view of the test point, control, or part, as shown in appendix 1. Enough of the major unit or chassis shall be shown to identify the location of the enlarged views.

(b) Leaders. - Except when a illustration relates to only one step, a leader or arrow shall point from each step number (see 3.3.3.2.3(a)) to clearly identify the equipment, part, or point involved in each step.

(c) Crossovers. - Test connections, leaders, arrows, or guidelines shall not cross, except where there is no alternative and clarity of intent does not suffer.

(d) Obstructions. - Step numbers, leaders, arrows, and guidelines shall be located so as not to obscure any pertinent portion of the illustration.

(e) Safety. - A WARNING note should appear at the beginning of each procedure where lethal voltage is involved (see NAVSHIPS 250-660-42).

3.3.3.2.3.4 Lubrication. - All necessary instructions and illustrations required for periodic lubrication of the equipment shall be included. Selection of the lubricant shall be in accordance with MIL-L-17192. The Military Specification of the specified lubricant shall be included in the Instructions.

3.3.3.2.3.5 Cleaning. - All necessary instructions, including the solvent (e.g., detergent), and illustrations required for periodic cleaning of specific equipment items, such as hidden filters, insulators, and electrical contacts, shall be included.

3.3.3.2.4 Recommended periodic-maintenance schedule. - All maintenance standards steps in the book recommended by the contractor to be performed on a periodic or routine basis for the purpose of preventive maintenance shall be listed. This list shall include all tests necessary for performance monitoring and preventive mechanical and electrical maintenance of the equipment or system. As shown in appendix 1, the list of steps shall be divided by the time periods for which the steps should be scheduled for accomplishment (i.e., daily, weekly, monthly, quarterly, semiannually, annually, or overhaul cycle, as applicable). Each step shall be identified by a step number and the preventive-maintenance "action required" (e.g., for monthly step 1, M1 - Test receiver sensitivity; and for annual step 2, A2 - Burnish antenna pedestal slip rings). In addition, each step shall refer to the location of the instructions for performing the step.

3.3.3.2.4.1 Scheduling periodic-maintenance steps. - These steps shall be scheduled on a routine basis (e.g., daily, weekly, monthly, etc, as applicable) except as noted in (a) and (b) below. The following factors and requirements shall be considered in preparing the schedule:

Note. - It should be understood that equipment will not be secured merely to perform these checks.

(a) Periodic maintenance shall be scheduled to keep the man-hour requirement to a minimum, consistent with effective preventive maintenance. Insofar as possible, checks or procedures should be

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T10:26Z
Check the source to verify that this is the current version before use.

designed so that they can be accomplished while the equipment is in use and while the operator is on watch.

(b) Daily checks or procedures shall be selected which will indicate a 90 percent confidence level of satisfactory equipment operation using built-in test features (e.g., test sets, meters, lamps, etc). Where this is impracticable because of lack of built-in test features, such checks shall be conducted on a weekly basis. Scheduled daily checks and procedures shall not, however, require performance time of more than 10 minutes for any single-unit set nor more than 30 minutes for multi-units. Any exceptions to this requirement shall be referred to the Bureau for approval.

(c) Test procedures where an equipment function or circuit is monitored shall be scheduled only if there is a definite likelihood that the interpretation of the test results will reveal potential trouble. Examples typical of this type of procedure are: tests of specific voltage distributions to critical circuits; tests for accuracy, such as counter circuit outputs or servo tracking tests; and tests for component deterioration, such as magnetron spectrum, ripple voltage, or shaft backlash tests.

(d) Tests should be scheduled on the basis of expected reliability of the circuit or function under test. For example, the hum level of an equipment may be expected to remain constant for a period of six months or longer, and should be scheduled accordingly.

(e) Preventive-maintenance procedures that cannot normally be performed when the ship is underway should be scheduled quarterly, or the instructions should indicate that the procedures are to be accomplished during the ship's in-port period (see 6.2 13). Examples typical of this type of procedure are cleaning and lubricating antennas, sonar beam pattern tests, sonar transducer maintenance, etc.

(f) Mechanical and electrical preventive maintenance procedures shall be scheduled for periodic accomplishment at intervals as infrequently as possible and yet guarantee against equipment failure due to lack of lubrication, insufficient ventilation, corrosion, or inadequate cleanliness.

(g) For equipments that are not designed to be used on a daily or continuous duty cycle (e.g., portable or emergency radio equipment, test equipment, equipment activated for training purposes, etc), the scheduling of the procedures shall be based on the number of hours of operation, or the minimum interval between which tests should be made to assure continued reliability (e.g., 100 hours of operation, 500 hours of operation, quarterly, etc).

3.3.4 Security classification. - The security classification of the book shall be as designated by the bureau or agency concerned. The Security Requirements Check List, DD Form 254, which constitutes a part of the contract for all classified material, indicates the classified features. All pages of classified books shall be marked in accordance with the Industrial Security Manual for Safe-guarding Classified Information, DD Form 441 (Attachment).

3.3.5 User activity comment sheet. - The maintenance standards book production

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T10:26Z
Check the source to verify that this is the current version before use.

source shall include one user activity comment sheet in each bound copy of the book. This sheet shall be located immediately following the last page of the book.

3.4 Quantity and distribution. - The quantity and distribution of Maintenance Standards Books shall be as specified in the contract, order, or ship specifications. Unless otherwise specified, the contractor shall be responsible for the distribution of all such books produced on the contract or order.

3.5 Revisions. - The method of revising books shall conform to the requirements of MIL-M-15071.

3.6 Reproduction rights. - Reproduction rights shall conform to the requirements of MIL-M-15071.

4. QUALITY ASSURANCE PROVISIONS

4.1 Inspection and tests. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for inspection of the book to ensure the fulfillment of all requirements as specified herein. The Government reserves the right to perform any inspections that are deemed necessary to assure that the books conform to the requirements of this specification.

4.2 Approval of review manuscripts. - Four copies of the review manuscript (see 6.2.14) which have been inspected in accordance with 4.1 shall be submitted to the bureau or agency concerned, for approval. Each copy shall have attached a filled-in quality-control contractor check-off list similar to that shown in appendix 2. Manuscripts shall be submitted as early as possible to ensure shipment of the printed books with the first production equipment. At least 30 days should be allowed for agency review and approval.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, and marking. - Packaging, packing, and marking shall be in accordance with MIL-M-15071.

5.1.1 Security. - Security shall be in accordance with the Industrial Security Manual for Safeguarding Classified Information, DD Form 441 (Attachment).

6. NOTES

6.1 Ordering data. - Procurement documents should specify the title, number, and date of this specification.

6.1.1 Quantity and distribution (See 3.4).

6.1.2 Inspection and tests (See 4.1).

6.2 Definitions.

6.2.1 POMSEE. - The term "POMSEE" is the abbreviation for Performance Operational and Maintenance Standards for Electronic Equipment.

6.2.2 Maintenance Standards Book. - A publication which includes reference standards tests, routine performance testing, and recommended preventive-maintenance instructions on specific serial-numbered equipments or systems, and which is used to implement POMSEE.

6.2.3 Maintenance standard. - A standardized method of testing and performing preventive maintenance on a particular type and model of electronic equipment or system.

6.2.4 Maintenance standard step. - A singularly identified testing or maintenance procedure in the Maintenance Standards Book.

6.2.5 Reference standard test. - A main-

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T10:26Z
Check the source to verify that this is the current version before use.

tenance standard procedure to be made, at the time of installation or equipment overhaul, on a properly operating equipment in order to provide technical personnel with a reference value (reference standard) that is a standard of normal performance or operation of an important equipment function or circuit.

6.2.6 Unscheduled reference standard test. - A reference standard test that is not to be made on a routine basis, but is included to provide a standard reference value in case a scheduled test indicates trouble, and to aid in isolating the trouble when troubleshooting the equipment or system (e.g., routine ac and dc voltage readings, etc).

6.2.7 Reference Standards Summary sheet. - A sheet (or sheets) preceding the front matter of a newly issued Maintenance Standards Book.

6.2.8 Performance standard. - A test used to evaluate the operational performance of the equipment or an equipment function, and which is tabulated in the Performance Standards Sheet for the equipment; for example, transmitter power output, minimum discernible signal, and source level measurement.

6.2.9 Performance Standards Sheet. - A sheet which lists the operational capability and the performance standards for all installations of a particular equipment or system. These sheets are prepared in accordance with MIL-S-21740 and are filed in the Binder for Electronic Equipment Performance Standard Sheet, NAVSHIPS 93000.

6.2.10 Recommended periodic-maintenance schedule. - A schedule in the Maintenance Standards Book to be used as an aid by the Electronics Officer in establishing a preventive-maintenance routine for a particular electronics equipment under his cognizance. The schedule indicates the maintenance standards steps that should be performed periodically, and the frequency of their performance, to prevent mechanical failure and to keep the equipment operating with minimum down time.

6.2.11 Test facilities. - Electrical and mechanical equipments, monitoring devices, special tools, jigs, and fixtures, either built-in or used separately but supplied with the equipment or system, which are necessary from both an operational and maintenance viewpoint to examine, test, monitor, measure, adjust, calibrate, or otherwise maintain the original functional operating status of an electronic system or equipment, as described in MIL-STD-415.

6.2.12 Test point. - An access point to a circuit, to be used so that a significant quantity can be measured or introduced to facilitate maintenance or monitoring.

6.2.13 In-port period. - The time during which a ship is in port or yard, or serviced by a tender, and is not activity engaged in operations or maintained in a condition of readiness.

6.2.14 Review manuscript. - Typewritten copy and rought artwork prepared to simulate the appearance of the final Maintenance Standards Book (including such details as running heads, running feet, and proper margins) and submitted to the bureau or agency concerned for review, comments, and approval.

6.2.15 Temporary change. - A temporary revision to the Maintenance Standards Book. It normally consists of instructions for making minor pen-and-ink corrections. It does not include revision pages. Temporary changes are normally replaced by a permanent change.

6.2.16 Permanent change. - A printed correction or revision for permanent in-

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T10:26Z
Check the source to verify that this is the current version before use.

corporation in a Maintenance Standards Book. It includes revised pages, but may also include instructions for making minor pen-and-ink corrections in the book. It supersedes temporary changes which have not been covered by a previous permanent change.

6.7.17 Field change. - An authorized modification or alteration made to an equipment after delivery to the Government.

6.3 Determination of tests. - Certain of the earlier POMSEE publications jeopardized the POMSEE program because they required unnecessary tests and, accordingly, more man-hours for accomplishment than were available to Navy maintenance personnel. After detailed analysis, contractors were required to revise original manuscripts, to delete unnecessary tests, or to relocate tests to a less frequent period of accomplishmend; for example, daily to weekly, weekly to monthly or quarterly, etc. Contractors are cautioned to avoid underestimating the time required to accomplish tests. (See 3.3.3.2.4.1.) Every effort should be made to ensure that the man-hours required are the minimum necessary for continued reliability.

6.4 Liaison with procuring agency. - Contractors who have never prepared Maintenance Standards Books or who desire in formulating the books, shall consult the bureau or agency concerned prior to commencing preparation.

Preparing activity:
Navy - Ships
(Project 7610-NO35SH)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T10:26Z
Check the source to verify that this is the current version before use.

Fold

Department of the Navy
Bureau of Ships
Washington, D.C. 20360

OFFICIAL BUSINESS

POSTAGE AND FEES PAID
NAVY DEPARTMENT

Chief, Bureau of Ships
Fleet Electronics Effectiveness Branch
Electronics Publications Section
Department of the Navy
Washington, D.C. 20360

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T10:26Z
Check the source to verify that this is the current version before use.

USER ACTIVITY TECHNICAL MANUAL COMMENT SHEET
NAVSHIPS 4914 (10-62)
(COG 1 - 11-DIGIT STOCK NUMBER: 0105-503-9850)

NAVSHIPS NO. _____

VOLUME NO. _____

(Fold on dotted line on reverse side, staple, and mail to BUSHIPS)

PROBLEM AREA:

Ship or Station_____

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T10:26Z
Check the source to verify that this is the current version before use.